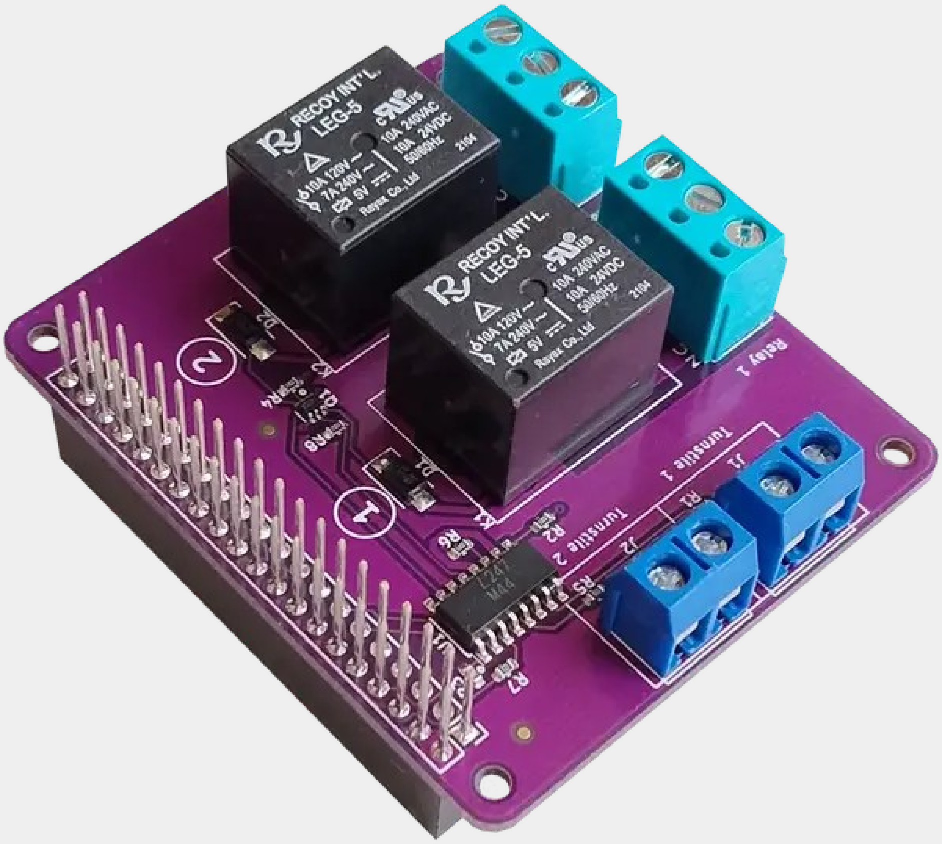
AQEX

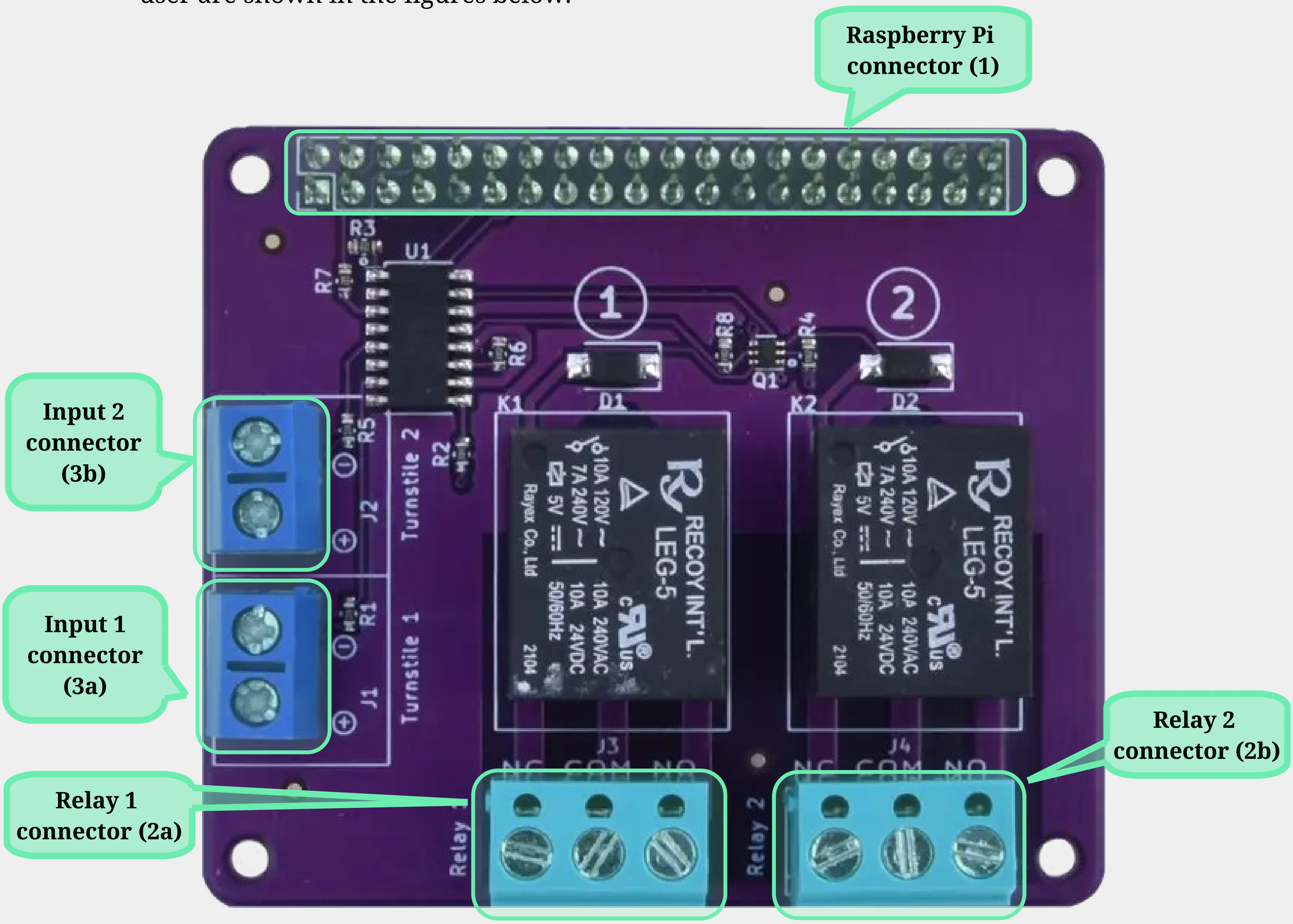


PIOHAT v1.0

User Manual v1.0
13/02/2026

Please read the instruction manual for safe use and a user experience tailored to your needs.

The PIOHAT has several connection, setup and feedback points for using it, highlighted in the figure below. For easy identification, the "( )" indicates the subsequent reference numbers. The colour and physical size of the connectors may differ from those shown in the figure below. The points of interest for the PIOHAT user are shown in the figures below.



PIOHAT v1.0 user points

1 Safety regulations

1.1 Personal safety

The PIOHAT has no replaceable parts - only the manufacturer or an accredited service centre can carry out repairs and maintenance.

1.2 Product safety

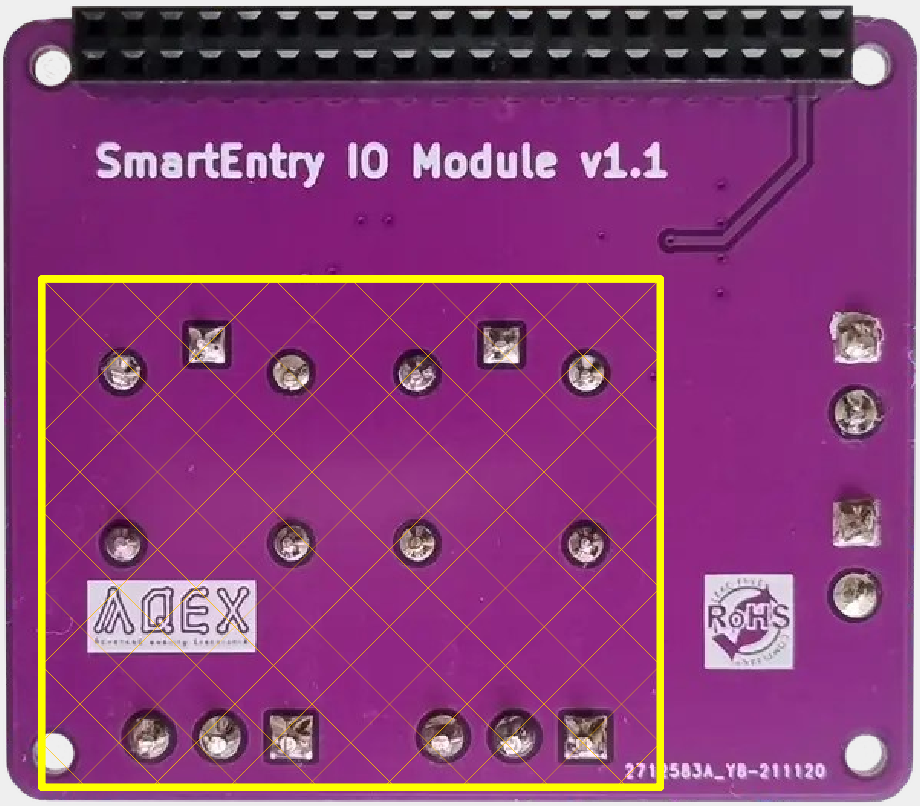
The PIOHAT product should be protected from too high or too low temperatures, direct sunlight. It should be kept in a dry place for 24 hours before installation.

Conductive liquids, plastic materials may cause short circuiting and permanent product damage, therefore avoid installation in such environments.

1.3 Precautions

The system operates from 5V, which is low voltage, so it is protected from electric shock in a life-safety sense. In case of short circuits caused by foreign matter, the contacts may heat up and cause injury!

The relays are suitable for switching voltages up to 250 VAC. If the device is used to control a high voltage device, touching the relay outputs or the (2) terminals is life-threatening. This section is marked with a yellow outline in the diagram below.



2 Introduction

Thank you for choosing the AQEX PIOHAT product to control your electronic devices and for the isolated detection of external digital signals!

The PIOHAT has been carefully and thoroughly designed to provide the most efficient way to ensure smooth operation in a wide range of conditions, to meet a wide range of user requirements.

By default, Raspberry Pi compatible microcomputers can only control devices with 3.3V DC, max. 16mA input current consumption via the GPIO port. With PIOHAT, they are also capable of switching higher voltages, current loads, and external electronic devices with alternating current (AC).

The GPIO ports of Raspberry Pi-compatible microcomputers can only directly accept 3.3V digital input signals.

The use of the PIOHAT module offers two advantages over a direct solution:

- Wider input signal level (3-45V)

- Isolation between input and GPIO port – use of optocoupler

3 Technical details

3.1 Inputs

The PIOHAT module has two input ports. Due to optical isolation, overvoltage and voltage spikes arriving at the input do not damage the Raspberry Pi.

The PIOHAT uses two GPIO pins on a 2x20 pin connector to monitor the inputs, the assignment of which is shown in Table 2.

GPIO BCM / BOARD pin (40 pin connector)	
Input (Turnstile) 1	Input (Turnstile) 2
GPIO17 / Pin 11	GPIO27 / Pin 13

Table 2 – Input ports

When using GPIO pins, make sure that the pins are correctly configured! All ports must be defined as inputs. Incorrect configuration may result in damage to the PIOHAT and/or the Raspberry Pi.

The logic signal (High/Low, 1/0, True/False) detected by the GPIO port is determined by the voltage difference between the two corresponding points of the input connector row (3). The GPIO port status is Low (0, False) below 3V and High (1, True) between 3 and 45V. The positive point of the input is ground-independent, while the negative point is connected to the PIOHAT and Raspberry Pi GND points.

Input levels above 45V will damage the input.

The input is polarity sensitive. Reverse polarity with a voltage level above 6V may damage the input. Reverse polarity below 6V will not damage the input, but no signal will be detected on the GPIO pin.

3.2 *Outputs*

Two relays are used for high-power switching, allowing two independent circuits to be controlled in parallel. The relay type is SPDT. The "COM" pin is the common point, which is connected to the "NC" pin in the default state and to the "NO" pin when energized.

Optocouplers between the GPIO ports and the relays isolate the two systems from each other, and the relay inputs are also completely isolated from the output circuit (double isolation).

The maximum voltage that can be switched by the relays under resistive load is 240V for alternating current (AC) and 24V for direct current (DC), and the maximum current that can be switched by the relays is 7A.

For inductive loads and 240V AC, the maximum current is 3A.

The PIOHAT uses 4 GPIO pins on the 2x20 pin connector for control, as shown in Table 3.

GPIO BCM / BOARD pin (40 pin connector)	
Relay 1	Relay 2
GPIO14 / Pin 8	GPIO15 / Pin 10

Table 3 – Relay controller ports

4 Commissioning

The PIOHAT product can be put into operation immediately after preparation and unpacking. For computers using a Raspberry PI-compatible 40-pin pin header, the connection is plug-and-play, while in other cases, the two-pin +/- connection or the 2+2 pins mentioned in Tables 2 and 3 can be used.

4.1 Power supply

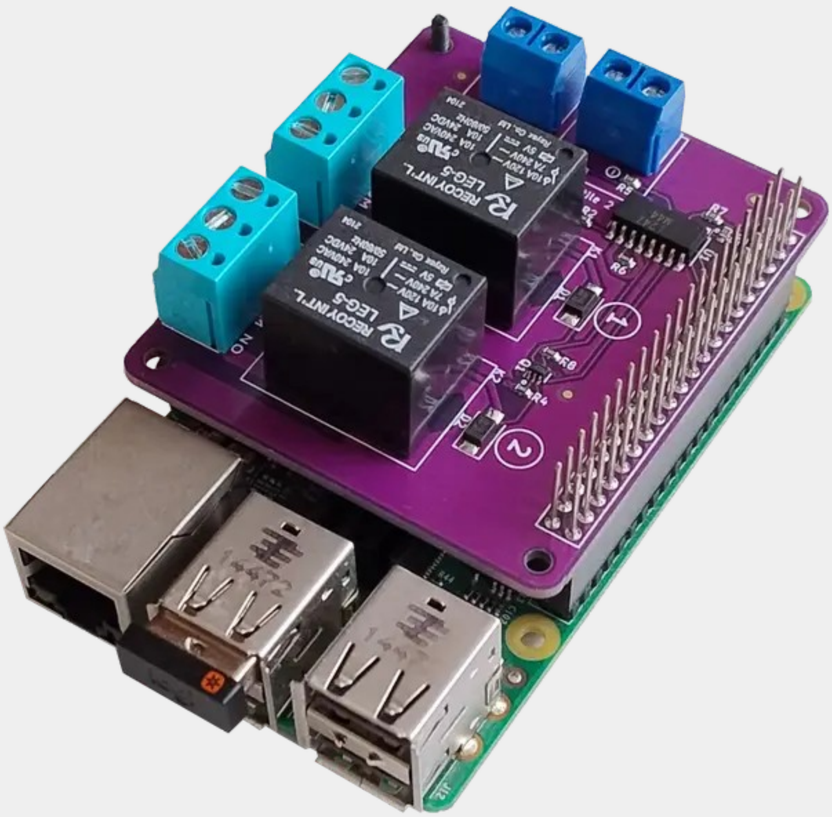
The PIOHAT is powered by the corresponding 5V legs of the 40 pin connector (1).

4.2 Connections

The controlling device can be connected to the PIOHAT product in different ways.

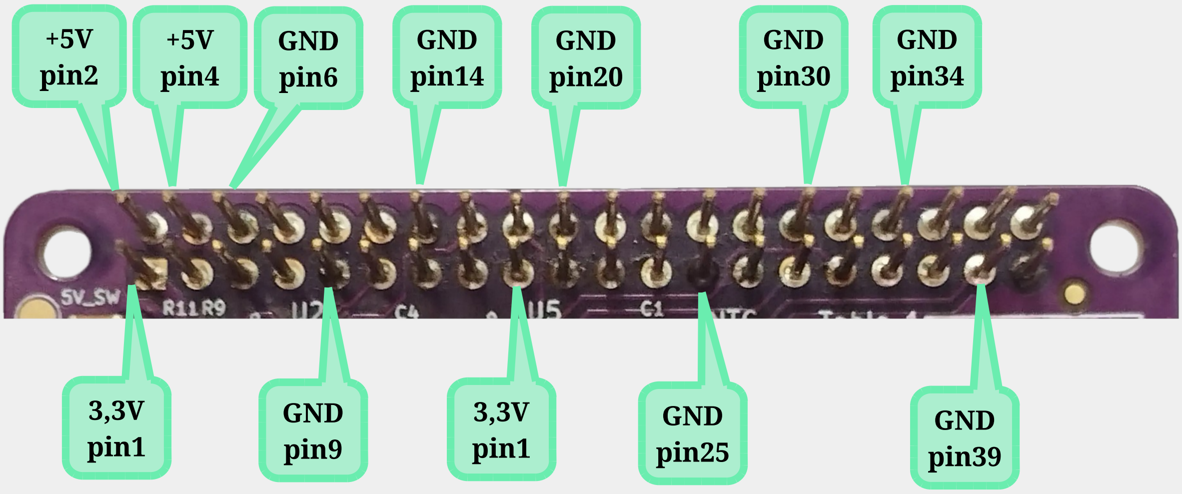
4.2.1 Single board computers (SBC)

If the SBC is equipped with a Raspberry PI compatible 40 pin header, the PIOHAT product is suitable for HAT connection. Simply plug the PIOHAT's connector (1) onto the computer's pin header as shown in the picture below.



4.2.2 Other devices

Any device, that can supply 5V DC, can be connected to the PIOHAT product using the appropriate contacts on the 40-pin header (1).



4.2.3 GPIO pins

The relays can be controlled via the pins mentioned in Table 3 of connector (1). The inputs are detected via the pins in Table 2 of connector (1).

All pins of the device connected via the 40-pin row (1) are also led out on the top side of the PIOHAT. With the exception of the 2+2 GPIO pins used by the PIOHAT, all pins can be freely used for any other purpose.

If you connect the PIOHAT 2x20 pin connector (1) to a non-Raspberry compatible device, please note the following:

- The relays can be controlled with 3.3V or 5V logic connected to the selected GPIO pin.

- In the case of inputs, the high levels on the corresponding pins of connector (1) will be determined by the voltage connected to the "3.3V" pin (pin1). The maximum value for this is 70V.

4.3 Setup

4.3.1 Hardware setup

The properly connected system will work under normal conditions with default settings.

4.3.2 Software setup

The operating system or program running on the control device can be configured to switch the relay outputs of the PIOHAT product and to detect the inputs via the appropriate GPIO connections.

Further information and C and shell-based utilities can be found at https://github.com/aqexhu/piohat

4.4 Troubleshooting

Symptom	Cause of error	Solution